

Aluminum 6063-O

Categories: [Metal](#); [Nonferrous Metal](#); [Aluminum Alloy](#); [6000 Series Aluminum Alloy](#)

Material Notes: Applications include pipe, railings, furniture, architectural extrusions, irrigation pipes, and transportation.

Data points with the AA note have been provided by the Aluminum Association, Inc. and are NOT FOR DESIGN.

Composition Notes:

Composition information provided by the Aluminum Association and is not for design.

Key Words: UNS A96063; ISO AlMg0.5Si; Aluminium 6063-O; AA6063-O; Al6063-O

Vendors: Visit [metalmen](#) for your metals needs. Products include special chemistry, tight tolerances, custom tempers, odd dimensions/forms, and small quantities. Phone 1-800-767-9494.

[Click here](#) to view all available suppliers for this material.

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	2.70 g/cc	0.0975 lb/in ³	AA; Typical
Mechanical Properties	Metric	English	Comments
Hardness, Brinell	25	25	AA; Typical; 500 g load; 10 mm ball
Tensile Strength, Ultimate	89.6 MPa	13000 psi	AA; Typical
Tensile Strength, Yield	48.3 MPa	7000 psi	AA; Typical
Modulus of Elasticity	68.9 GPa	10000 ksi	AA; Typical; Average of tension and compression. Compression modulus is about 2% greater than tensile modulus.
Poissons Ratio	0.33	0.33	
Fatigue Strength	55.2 MPa @# of Cycles 5.00e+8	8000 psi @# of Cycles 5.00e+8	completely reversed stress; RR Moore machine/specimen
Machinability	30 %	30 %	0-100 Scale of Aluminum Alloys
Shear Modulus	25.8 GPa	3740 ksi	
Shear Strength	68.9 MPa	10000 psi	AA; Typical
Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.00000299 ohm-cm @Temperature 20.0 °C	0.00000299 ohm-cm @Temperature 68.0 °F	AA; Typical
Thermal Properties	Metric	English	Comments
CTE, linear 	21.8 µm/m-°C @Temperature -50.0 - 20.0 °C	12.1 µin/in-°F @Temperature -58.0 - 68.0 °F	
	23.4 µm/m-°C @Temperature 20.0 - 100 °C	13.0 µin/in-°F @Temperature 68.0 - 212 °F	AA; Typical; average over range
	24.5 µm/m-°C @Temperature 20.0 - 200 °C	13.6 µin/in-°F @Temperature 68.0 - 392 °F	
	25.6 µm/m-°C @Temperature 20.0 - 300 °C	14.2 µin/in-°F @Temperature 68.0 - 572 °F	
	Specific Heat Capacity 0.900 J/g-°C	0.215 BTU/lb-°F	
	Thermal Conductivity 218 W/m-K	1510 BTU-in/hr-ft ² -°F	AA; Typical at 77°F
Melting Point	616 - 654 °C	1140 - 1210 °F	AA; Typical range based on typical composition for wrought products >= 1/4 in. thickness
Solidus	616 °C	1140 °F	AA; Typical
Liquidus	654 °C	1210 °F	AA; Typical
Processing Properties	Metric	English	Comments
Annealing Temperature	413 °C	775 °F	hold at temperature for 2 to 3 hr; cool at 50°F per hour from 775 to 500°F
Solution Temperature	521 °C	970 °F	
Aging Temperature	177 °C	350 °F	hold at temperature for 8 hr

Component Elements Properties	Metric	English	Comments
Aluminum, Al	<= 97.5 %	<= 97.5 %	As remainder
Chromium, Cr	<= 0.10 %	<= 0.10 %	
Copper, Cu	<= 0.10 %	<= 0.10 %	
Iron, Fe	<= 0.35 %	<= 0.35 %	
Magnesium, Mg	0.45 - 0.90 %	0.45 - 0.90 %	
Manganese, Mn	<= 0.10 %	<= 0.10 %	
Other, each	<= 0.05 %	<= 0.05 %	
Other, total	<= 0.15 %	<= 0.15 %	
Silicon, Si	0.20 - 0.60 %	0.20 - 0.60 %	
Titanium, Ti	<= 0.10 %	<= 0.10 %	
Zinc, Zn	<= 0.10 %	<= 0.10 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.